

Roll No.

TRA-501

B. TECH. (ME-RA) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
ROBOTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define the terms robot and robotics. What are different laws of robotics ?
(CO1)

OR

- (b) How to define the degrees of freedom of a robot manipulator ? Also, define the following types of manipulators : Spatial manipulator, Planar manipulator, and Redundant manipulator.
(CO1)

2. (a) Discuss about the various robot arm configurations with neat sketches.
(CO1)

OR

- (b) What are different components of a robot as a complete robotic system ? Discuss each component briefly.
(CO1)

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(2)

3. (a) Discuss about the mapping in general frames and describe the homogenous transform. (CO2)

OR

- (b) How to evaluate the vector transformation considering the vector translation and rotation with respect to a given frame ? (CO2)
4. (a) A frame {B} is rotated relative to another frame {A} about Z axis by 60 degrees and translated 5 units in X and 7 units in Y. The Z axis is pointing out of the page. Thus, we have a description of ${}^A_B T$. Find ${}^B_A T$. (CO2)

OR

- (b) A frame {B} is rotated relative to a frame {A} about z axis by 45 degrees; the z axis is pointing out of the page. Position vector of a point with respect to frame {B} is given by $[3 \ 2 \ 0]^T$. Calculate the position of that point with respect of frame {A}. (CO2)
5. (a) What do you understand by the D-H notation ? Write the different steps involved in D-H algorithm to derive the forward kinematic model. (CO3)

OR

- (b) What are differences between the forward and the inverse kinematic models ? Also, briefly describe the joint-link parameters of a manipulator. (CO3)